

Title: Crop Disease Detection Using Image and Relational Data

1. ABSTRACT

Agriculture is the backbone of many economies, and ensuring the health of crops is vital for sustainable food production. Plant diseases represent a significant threat to crop yield, farmer income, and food security. Early detection and timely intervention can mitigate these risks, but access to expert diagnostics is often limited in rural and underdeveloped regions. To address this issue, our project introduces a hybrid approach that combines deep learning with structured relational data to detect crop diseases from leaf images.

We use a Convolutional Neural Network (CNN) based on the ResNet9 architecture to classify images into healthy or diseased categories. Simultaneously, we develop an end-to-end Extract, Transform, Load (ETL) pipeline that converts unstructured image data into structured tabular data suitable for storage in a relational database. This structured metadata is further analyzed and visualized using Power BI to extract actionable insights on disease distribution and severity. Our system demonstrates how artificial intelligence (AI), data engineering, and visualization tools can be integrated to form a decision-support system for modern agriculture. This solution not only enhances disease detection capabilities but also promotes data-driven farming practices that are scalable, interpretable, and efficient.

2. INTRODUCTION

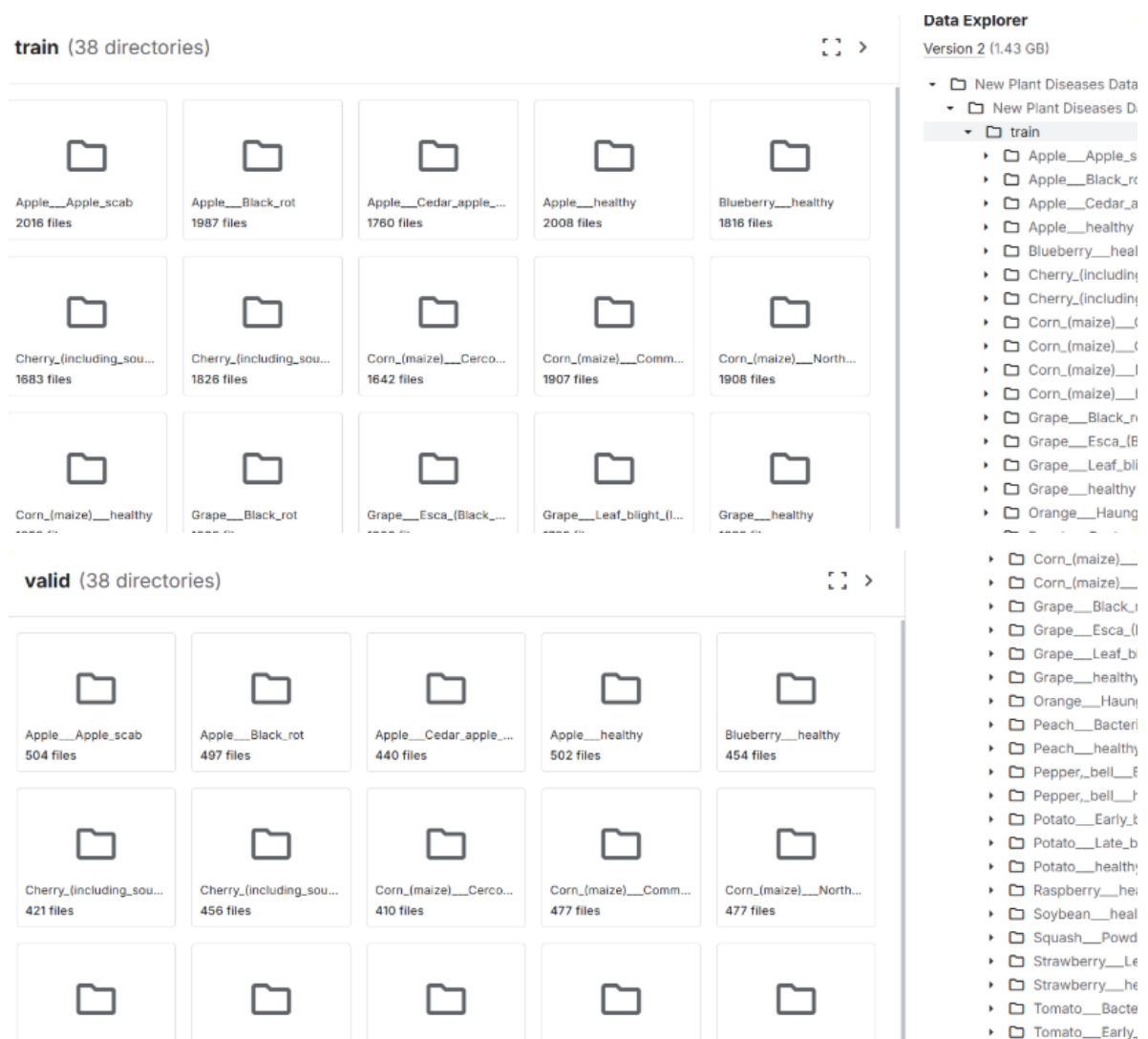
In the modern era, agriculture continues to face persistent challenges, one of the most critical being crop diseases. These diseases not only diminish yield but also threaten food supply chains and farmer livelihoods. Traditional methods of diagnosing plant diseases rely heavily on expert observation and laboratory analysis, both of which can be time-consuming, expensive, and inaccessible to many farmers. The lack of scalable diagnostic tools results in delayed interventions, leading to larger outbreaks and crop losses.

In response to this issue, our project explores an innovative solution: a smart, automated system that uses artificial intelligence to detect plant diseases based on leaf images. The main goal of this project is to create a reliable, user-friendly platform that helps in early disease identification, reduces the need for manual diagnosis, and promotes precision

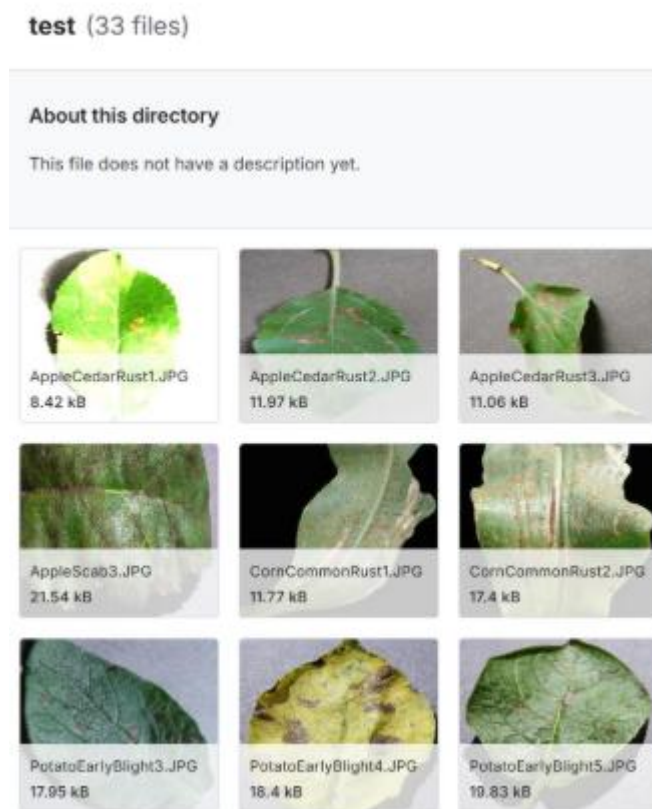
agriculture. Our approach not only focuses on the application of deep learning models for classification tasks but also integrates data engineering techniques to transform image data into structured formats for storage, querying, and analysis. This combination of computer vision and relational data analysis offers a powerful framework for tackling agricultural problems in a real-world, scalable manner.

A. Dataset Description

To develop and evaluate our disease detection model, we utilized a publicly available dataset from Kaggle containing approximately 87,000 RGB images of crop leaves. These images are organized into 38 distinct categories, with each category representing either a healthy leaf or a specific plant disease. The dataset covers a wide range of crops, such as corn, tomato, grape, bell pepper, and more, providing a comprehensive basis for training our classification model. The dataset is initially unstructured, stored in folders where the folder names encode the crop type and disease label.



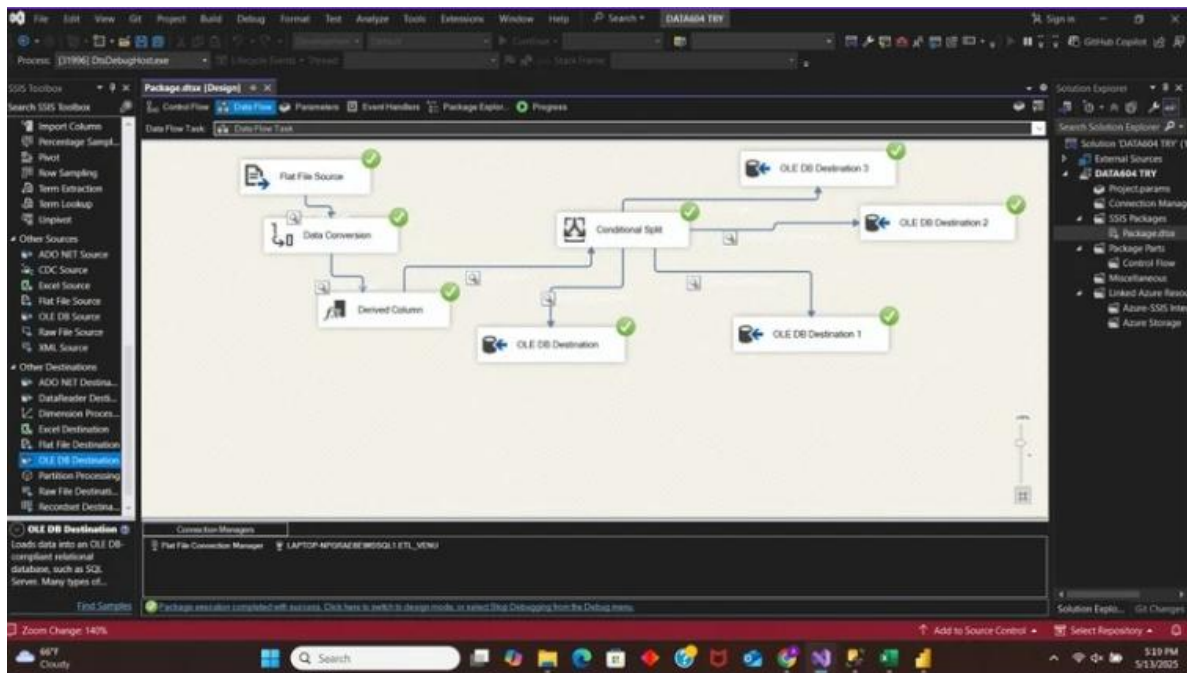
For training and validation purposes, we split the data into an 80:20 ratio. An additional test set of 33 unseen images was preserved to evaluate the model's performance in real-world scenarios. While the primary data consists of images, we extracted essential metadata — including image filenames, crop types, and disease labels — to construct a structured dataset. This structured data allows us to go beyond simple classification and perform advanced querying, reporting, and visualization.



By using tools like SSIS (SQL Server Integration Services), we were able to automate the creation of relational tables and prepare the data for further analysis using Power BI. This data-centric approach strengthens the analytical capabilities of our system and provides rich contextual insights.

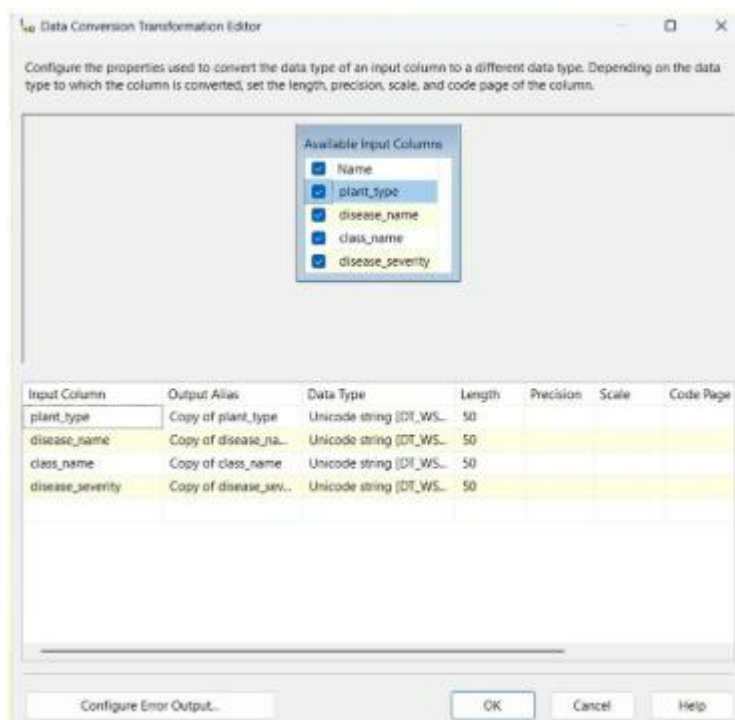
3. ETL Pipeline: Extract, Transform, Load

In data-driven systems, the ETL pipeline plays a critical role in moving data from raw, unorganized formats to structured, analysis-ready forms. Our pipeline emulates enterprise-level data workflows using tools such as Python and SSIS.



i. Extraction Phase

The extraction process involves collecting raw image data from a directory of folders, where each folder is named according to the crop type and disease condition (e.g., Tomato__Late_blight). This hierarchical folder structure provides useful metadata embedded within the folder names. All 87,000 images are initially extracted from these folders using Python scripts. Each image is an RGB file with a 256x256 resolution, ensuring standardization for subsequent processing.

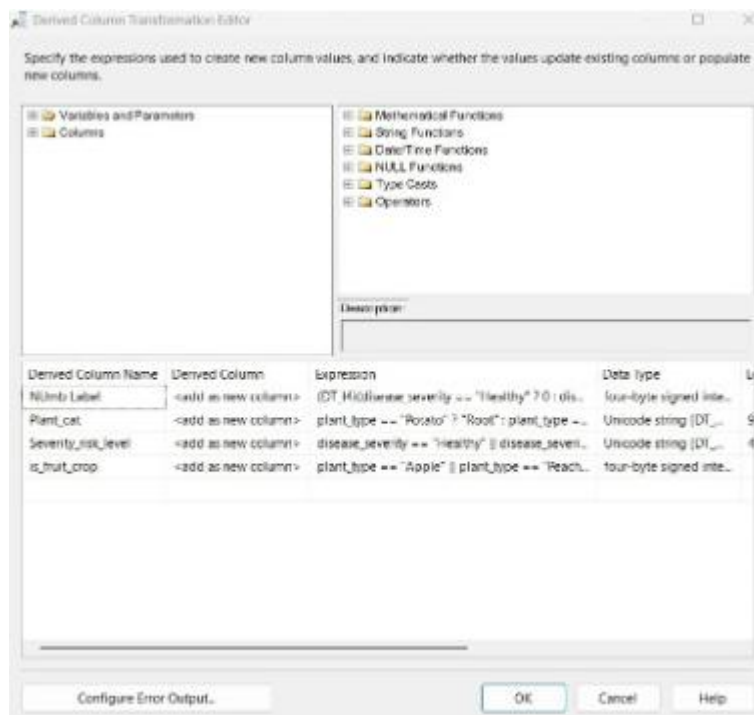


ii. Transformation Phase

This phase is where we convert unstructured data (images) into structured formats. We use Python to extract metadata — such as the crop name, disease type, and file name — from the folder and file paths. This metadata is then assembled into a tabular format using Pandas, resulting in a DataFrame with columns like:

- Crop_Type
- Disease_Name
- Image_Filename
- File_Path

Data cleaning and normalization are also part of this step to ensure consistency in naming conventions (e.g., unifying Bell_Pepper and bellpepper into a standard format).



iii. Loading Phase

After transformation, the structured data is exported to a relational database using SQL Server. This step involves defining appropriate table schemas and using SQL Server Integration Services (SSIS) to automate the data loading process. The database design includes normalization to eliminate redundancy and support efficient querying. The loaded data can now be used for analytics, reporting, and visualization.



iv. Integration with Other Tools

We integrated the database with visualization tools such as Power BI and also connected it with the CNN model's prediction output. This integration allows the entire data pipeline — from raw image to actionable insights — to function seamlessly.

4. Deep Learning Model: ResNet9-Based CNN

To perform disease classification on leaf images, we employed a Convolutional Neural Network (CNN) based on the ResNet9 architecture. This model is optimized for image classification tasks with a balance of speed and accuracy.

4 Why ResNet9?

We selected ResNet9 because it includes residual connections that help the network learn more efficiently. Unlike deeper models that are harder to train, ResNet9 is lightweight but still powerful, making it suitable for resource-constrained environments. Its architecture is well-suited to our dataset size and complexity.

4.2 Model Architecture

The architecture includes the following layers:

Input Layer: Accepts 256x256 RGB images.

Convolutional Blocks: Several Conv2D layers are used to extract hierarchical features like edges, textures, and leaf discoloration patterns.

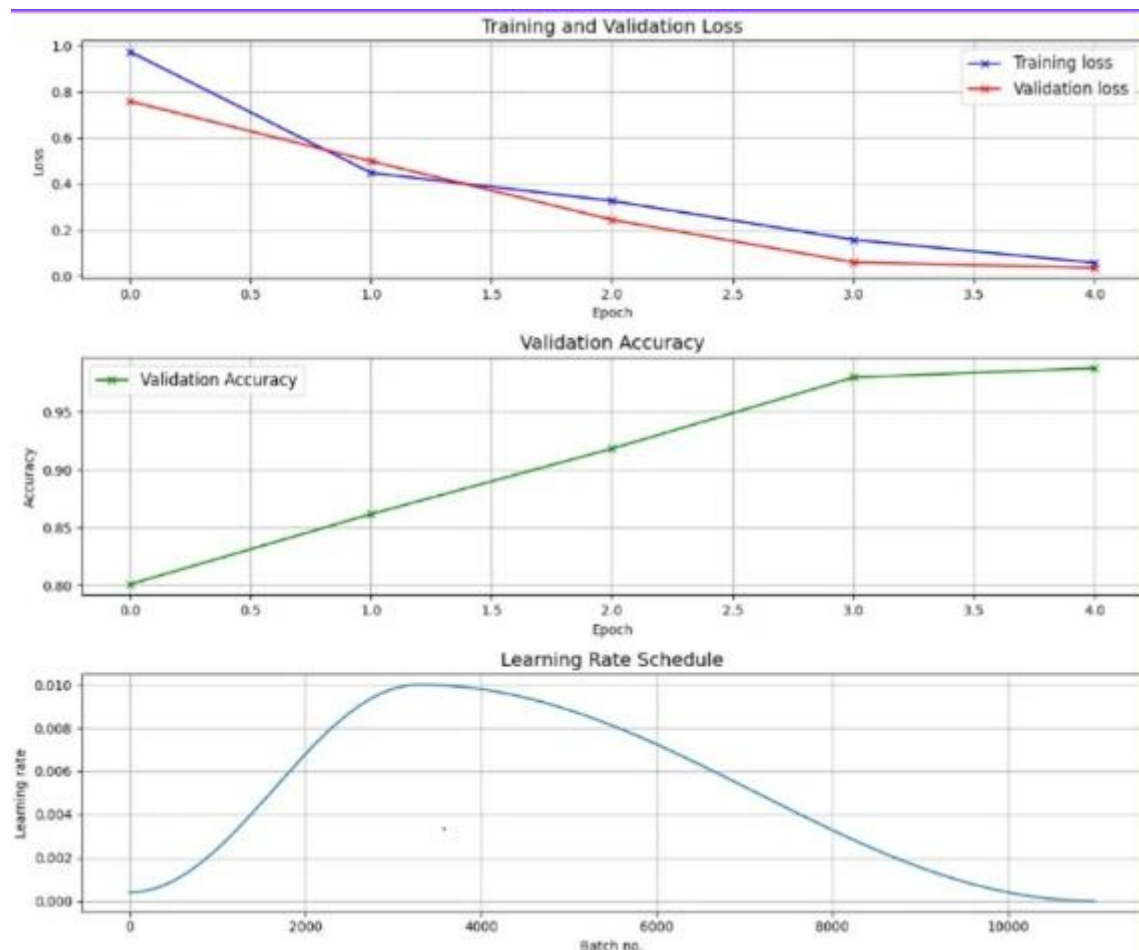
Residual Blocks: Two residual connections allow the model to “skip” over certain layers, enabling better gradient flow and faster convergence.

Pooling Layers: MaxPooling layers reduce spatial dimensions while retaining important features.

Fully Connected Layer: The final Dense layer maps the extracted features to 38 output categories.

Activation: Uses ReLU for hidden layers and softmax for the output.

Validation:



4.3 Model Training

The model was trained using the following configurations:

Loss Function: Categorical Cross-Entropy

Optimizer: Adam

Epochs: 25–30

Batch Size: 32

Data Augmentation: Used ImageDataGenerator for rotation, flipping, and zooming to prevent overfitting.

Regularization: Dropout and Batch Normalization layers were added to enhance generalization.

4.4 Model Evaluation

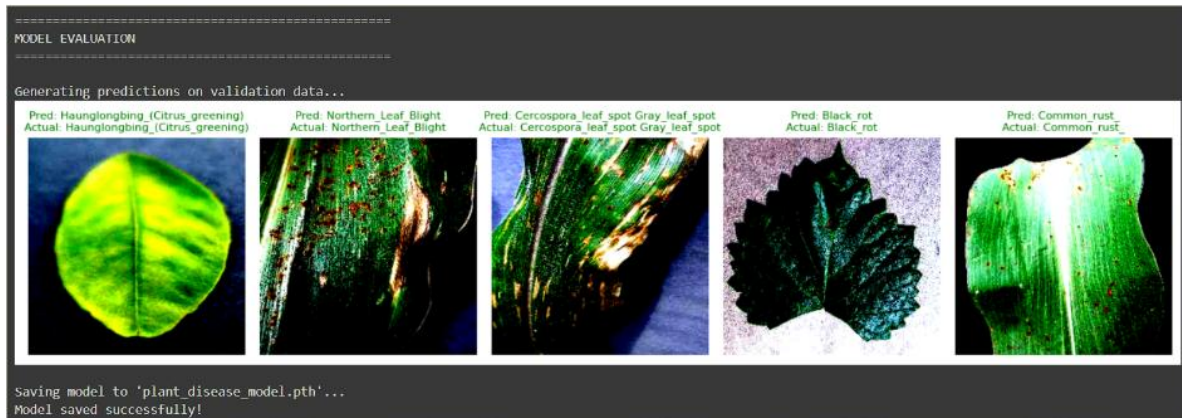
The model was evaluated on a held-out test set of 33 unseen images. Metrics such as accuracy, precision, recall, and F1-score were used. The model achieved high accuracy and generalizability, correctly classifying images even from underrepresented classes.

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Training for 2 epochs with:
- Maximum learning rate: 0.01
- Weight decay: 0.0001
- Gradient clipping: 0.1
- Optimizer: Adam

Starting training...

Epoch 1/2
Training:
Train Epoch 1: 100% ██████████ 2197/2197 [15:08<00:00, 2.44it/s, loss=0.875]
Validating:
Valid Epoch 1: 100% ██████████ 550/550 [01:14<00:00, 7.75it/s, acc=1, loss=0.0261]
Epoch [0], last_lr: 0.00812, train_loss: 0.8748, val_loss: 0.4814, val_acc: 0.8509
Epoch 1 Summary:
  Training Loss: 0.8748
  Validation Loss: 0.4814
  Validation Accuracy: 85.09%

Epoch 2/2
Training:
Train Epoch 2: 100% ██████████ 2197/2197 [15:10<00:00, 2.46it/s, loss=0.148]
Validating:
Valid Epoch 2: 100% ██████████ 550/550 [01:14<00:00, 7.67it/s, acc=1, loss=9.94e-5]
Epoch [1], last_lr: 0.00000, train_loss: 0.1480, val_loss: 0.0291, val_acc: 0.9909
Epoch 2 Summary:
  Training Loss: 0.1480
  Validation Loss: 0.0291
  Validation Accuracy: 99.09%
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5. Power BI Visualization and Analysis

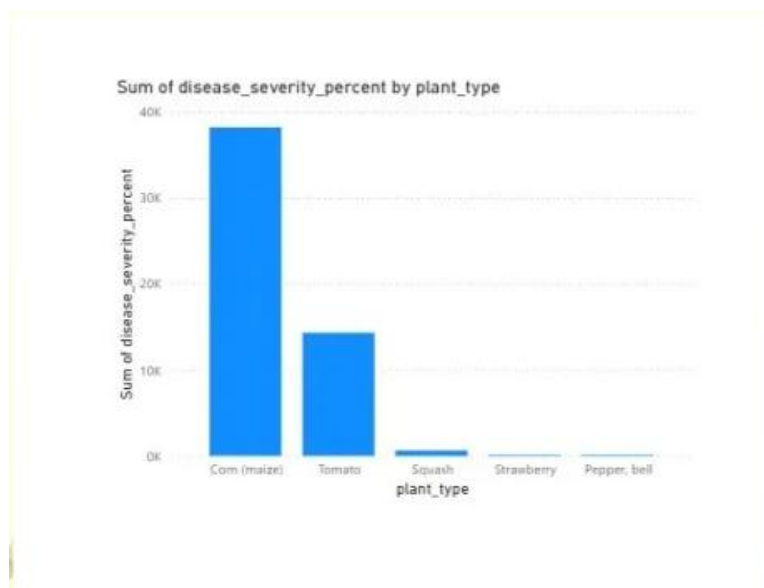
Power BI plays a central role in transforming our structured data into meaningful visual insights that can guide decision-making.

5.1 Dashboard Overview

We developed interactive dashboards that connect to the SQL Server database. These dashboards allow users to:

- View disease distribution by crop type
- Monitor severity levels
- Analyze model predictions
- Compare healthy vs diseased samples

Power BI Analysis of Disease Severity



5.2 Disease Distribution Across Crops

One of the main visualizations is a bar chart showing how many samples of each disease were detected per crop. This helps identify high-risk crops like Tomato and Corn, which showed the highest number of diseased samples.

5.3 Severity Analysis

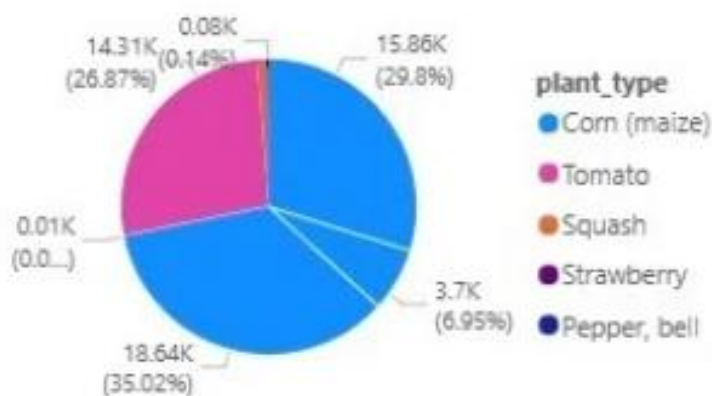
The severity bar chart aggregates the severity scores or image counts by crop type. Corn was shown to be the most severely affected, followed by Tomato. This analysis supports prioritizing resources for those crops.

5.4 Pie Chart Analysis

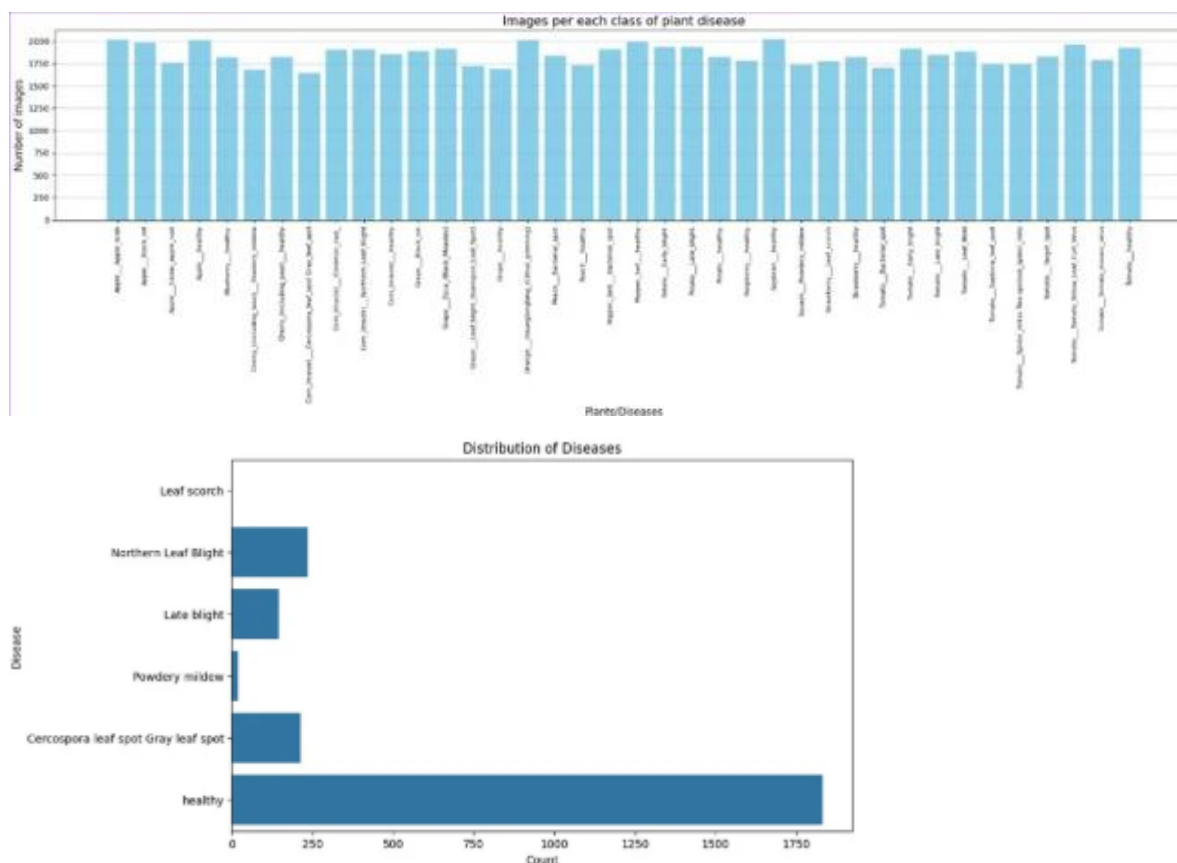
A pie chart was created to show the percentage contribution of each disease and crop type to the total disease burden. This dual-layer visualization helps users understand the overlap between crop susceptibility and disease spread.

Power BI Analysis of Disease Severity:

Sum of disease_severity_percent by plant_type and disease_name



Data Visualization Using Python



5.5 Class Balance and Prediction Metrics

Additional visuals include class distribution charts, confusion matrices, and prediction accuracy per class. These are essential for evaluating model fairness and diagnosing potential biases.

6. Real-World Application and Deployment Strategy

This project is designed with real-world usability in mind, mimicking data workflows used in smart farming systems.

6.1 Target Users

Farmers: Can capture and upload images using smartphones or IoT cameras.

Agronomists: Use dashboards for disease monitoring and forecasting.

Researchers: Analyze historical disease data and test agricultural hypotheses.

Policymakers: Monitor crop health at a regional or national level.

6.2 Deployment Options

The solution can be deployed on:

Mobile platforms using lightweight CNNs and cloud inference

Edge devices like Raspberry Pi for in-field processing

Cloud services for scalable model training and storage

6.3 Hardware and Software Requirements

Image Capture: Smartphone or drone

Processing Hardware: GPU-enabled machine or Raspberry Pi with USB camera

Software Stack: Python (TensorFlow, Pandas)

- SQL Server
- SSIS
- Power BI

6.4 Cost Estimate

The prototype version of this system can be developed and deployed with an estimated budget between \$12,000 to \$18,000 USD. This includes:

- Basic hardware setup
- Software development
- Cloud services for hosting and storage
- Field testing and pilot deployment

6.5 Scalability and Future Enhancements

The current system is scalable. Future enhancements could include:

- Adding weather and soil data
- Real-time alerts
- Multilingual user interfaces
- Integration with government crop monitoring schemes

Team Members:

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